

Skill-02 Task 02 - Keyboard Input Handling

Objective

To capture keyboard input, handle Backspace, process the Enter key, manage an input buffer, support multi-character commands, and test user interaction.

Features

- Character-by-character keyboard input
- Enter key processing
- Backspace handling
- Input buffer management
- Multi-character command support
- Exit command
- Interactive user testing

Source File

keyboard_input.c

Compilation

```
gcc keyboard_input.c -o keyboard_input
```

Execution

```
./keyboard_input
```

Sample Interaction

```
Shellforge> hello Command entered: hello
```

```
Shellforge> testing Command entered: testing
```

```
Shellforge> exit Exiting Shellforge...
```

Result

Keyboard input handling was successfully implemented and tested.

```
#include <stdio.h>

#include <unistd.h>

#include <termios.h>


#define BUFFER_SIZE 100


int main()
{
    struct termios oldt, newt;

    char buffer[BUFFER_SIZE];

    int position = 0;

    char ch;


    tcgetattr(STDIN_FILENO, &oldt);

    newt = oldt;

    newt.c_lflag &= ~(ICANON | ECHO);

    tcsetattr(STDIN_FILENO, TCSANOW, &newt);

    printf("Shellforge> ");

    fflush(stdout);

    while (1)
    {
        ch = getchar();

        if (ch == '\n')
```

```
{  
    buffer[position] = '\0';  
  
    printf("\nCommand entered: %s\n", buffer);  
  
    if (position == 4 &&  
        buffer[0] == 'e' &&  
        buffer[1] == 'x' &&  
        buffer[2] == 'i' &&  
        buffer[3] == 't')  
    {  
        break;  
    }  
  
    position = 0;  
  
    printf("Shellforge> ");  
    fflush(stdout);  
}  
else if (ch == 127 || ch == 8)  
{  
    if (position > 0)  
    {  
        position--;  
  
        printf("\b \b");  
        fflush(stdout);  
    }  
}
```

```
    }  
    else if (position < BUFFER_SIZE - 1)  
    {  
        buffer[position] = ch;  
        position++;  
  
        putchar(ch);  
        fflush(stdout);  
    }  
}  
  
tcsetattr(STDIN_FILENO, TCSANOW, &oldt);  
  
printf("Exiting Shellforge...\n");  
  
return 0;  
}
```

Output-

Shellforge> hello

Command entered: hello

Shellforge> testing

Command entered: testing

Backspace Test:

Shellforge> helo

After pressing Backspace and entering l:

hello

Command entered: hello

Shellforge> exit

Exiting Shellforge...